Gono Bishwabidyalay

Department of Computer Science & Engineering

Mid-Term Examination: July- December 2026

Program: B. Sc. (Hons.) in CSE, Semester: 1st

Course Code: MAT 1104 **Course Title: Math-I (Calculus)**

Full marks: 20 **Time: 60 minute**

[N. B: Answer any Two questions. Figures in the right margin indicate marks]

1. a) Define monotonically increasing and decreasing function with example. Find the domain and range of the function $f(x)=\frac{1}{2}\left(x+|x|\right)$. [6]

b) Draw the graph of the following function $f(x)=\begin{cases} -1 \text{ when } x\le -1 \\ x \text{ when } -1<x<1 \\ 1 \text{ when } x\ge 1 \end{cases}$. [4]

2. a) Define limit of a function. Prove that $\lim_{x\to 2}(3x+4)=10$, applying (ε,δ) definition of limit. [5]

b) Define continuity. A function $f(x)$ is defined as follows [5]

$$f(x)=\begin{cases} 1 & when \quad x<0 \\ 1+\sin x & when \quad 0<x<\frac{\pi}{2} \\ 2+\left(x-\frac{\pi}{2}\right)^2 & when \quad x\ge\frac{\pi}{2} \end{cases}$$ Discuss the continuity at $x=\frac{\pi}{2}$.

3. a) Differentiate $\tan^{-1}\frac{\sqrt{1+x^2}-1}{x}$ with respect to $\tan^{-1}x$.. [4]

b) If $f(x)=\left(\frac{a+x}{b+x}\right)^{a+b+2x}$ show that $f'(0)=\left\{2\log\left(\frac{a}{b}\right)+\frac{b^2-a^2}{ab}\right\}\left(\frac{a}{b}\right)^{a+b}$. [6]